

BALAKRISHNA NAIR NALIVALAPIL

📍 College Park, MD 📞 (240) 476-6045 ✉ bkrishna@umd.edu
🌐 [linkedin.com/in/bkrishnanair](https://www.linkedin.com/in/bkrishnanair) 🐙 github.com/bkrishnanair 🌐 bkrishnanair.github.io

PROFESSIONAL SUMMARY

Software Engineer combining 2 years of enterprise backend experience with founder-level product ownership, holding a Master's in Software Engineering from the University of Maryland (May 2026). Shipped RESTful services and integration pipelines within a 100+ member engineering organization processing \$100M+ daily, and independently built and launched a production geospatial platform serving live users. Brings enterprise CI/CD discipline and startup shipping velocity across Java, Python, TypeScript, and cloud-native stacks.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C++

Backend & Systems: REST APIs, Microservices, Event-Driven Architecture (WebSphere MQ), Spring Boot, Node.js, Django, Distributed Systems, Data Structures & Algorithms

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), GCP, Firebase, Docker, Jenkins, JFrog Artifactory, CI/CD, Git, GitHub Actions, Linux/Unix

Databases: PostgreSQL, MongoDB, Firestore, MySQL, Redis

Frontend: React, Next.js, Tailwind CSS, HTML5/CSS3

AI/ML: Generative AI integration (Gemini, Llama3), Prompt Engineering, Pandas

PROFESSIONAL EXPERIENCE

Capgemini Technology Services

Oct 2022 – Sep 2024

Software Engineer → Senior Software Engineer, Enterprise Integration

Chennai, India

- Delivered 20+ production RESTful services and integration pipelines within a 100+ member global engineering organization supporting MUFG Bank's transaction platform processing \$100M+ daily, by designing Java-based middleware on IBM Integration Bus.
- Reduced post-deployment incidents 25% by building event-driven middleware with structured logging and monitoring across distributed services using WebSphere MQ.
- Cut feature integration time 30% across 5 application teams by architecting automated build-test-deploy CI/CD pipelines with Jenkins and JFrog Artifactory.
- Guaranteed data integrity across heterogeneous banking systems by designing validation and transformation layers handling JSON/XML payloads at high message volume.
- Reduced onboarding ramp-up 15% by authoring architectural decision records and mentoring 3 junior engineers; promoted to Senior Software Engineer within 18 months.

PROJECTS

Huddle | Creator & Lead Engineer | Next.js, TypeScript, React, Firebase, Google Maps API

Aug 2025 – Present

- Launched a production hyperlocal event-discovery platform with live campus users by building the full stack solo — Next.js 15, TypeScript, and Firebase Admin SDK authentication with HTTP-only cookie sessions.
- Prevented race conditions across 500+ events under concurrent load by engineering transactional Firestore writes for RSVPs and event state.
- Cut redundant API calls 90% and achieved 60fps mobile rendering by implementing viewport debouncing and memoized React marker components.
- Selected for UMD's Pitch Dingman Competition (Quattrone Tech Track); deployed on Vercel with automated CI/CD preview pipelines.

ClassNest – AI-Powered LMS | Django, React, PostgreSQL, Llama3, Docker

Sep 2024 – Nov 2024

- Enforced 90% test coverage via Test-Driven Development while building a microservices-ready LMS with Django and React.
- Automated PDF-to-quiz conversion by engineering an LLM pipeline with Llama3 via Groq API; containerized with Docker and shipped through CI/CD.

IoT Smart Health Monitoring System | Python, Arduino, Raspberry Pi, Sensors

2021 – 2022

- Enabled remote patient vitals tracking with threshold-based alerts by integrating 5 hardware sensors into a real-time monitoring dashboard.

EDUCATION

University of Maryland, College Park

May 2026

Master of Engineering in Software Engineering — GPA: 3.7/4.0

- **Coursework:** Cloud Computing, Cloud Security, Software Design & Architecture, Virtualization & Containers, Software Project Management

Indian Institute of Information Technology, Sri City

May 2022

Bachelor of Technology in Electronics & Communications Engineering

ACHIEVEMENTS & CERTIFICATIONS

- Selected for the **UMD Pitch Dingman Competition** (Quattrone Tech Track) for Huddle
- **AWS Certified Solutions Architect – Associate** (SAA-C03) — in progress
- Promoted from Software Engineer to Senior Software Engineer at Capgemini within 18 months